

Step 1: Import Libraries

```
import pandas as pd
import matplotlib.pyplot as plt
```

Insight:

Pandas is used for data manipulation and Matplotlib is used for visualization.

Step 2: Load Dataset

```
df = pd.read_csv("amazon.csv")
```

Insight:

Dataset successfully loaded into DataFrame.

Step 3: Understand Data

```
df.head()
```

	product_id	product_name \
0	B07JW9H4J1	Wayona Nylon Braided USB to Lightning Fast Cha...
1	B098NS6PVG	Ambrane Unbreakable 60W / 3A Fast Charging 1.5...
2	B096MSW6CT	Sounce Fast Phone Charging Cable & Data Sync U...
3	B08HDJ86NZ	boAt Deuce USB 300 2 in 1 Type-C & Micro USB S...
4	B08CF3B7N1	Portronics Konnect L 1.2M Fast Charging 3A 8 P...

	category	discounted_price
0	Computers&Accessories Accessories&Peripherals ...	₹399
1	Computers&Accessories Accessories&Peripherals ...	₹199
2	Computers&Accessories Accessories&Peripherals ...	₹199
3	Computers&Accessories Accessories&Peripherals ...	₹329
4	Computers&Accessories Accessories&Peripherals ...	₹154

	actual_price	discount_percentage	rating	rating_count \
0	₹1,099	64%	4.2	24,269
1	₹349	43%	4.0	43,994
2	₹1,899	90%	3.9	7,928
3	₹699	53%	4.2	94,363
4	₹399	61%	4.2	16,905

	about_product \
0	High Compatibility : Compatible With iPhone 12...

1 Compatible with all Type C enabled devices, be...
2 【Fast Charger& Data Sync】-With built-in safet...
3 The boAt Deuce USB 300 2 in 1 cable is compati...
4 [CHARGE & SYNC FUNCTION]- This cable comes wit...

user_id \

0 AG3D604STAQKAY2UVGEUV46KN35Q,AHMY5CWJMMK5BJRBB...
1 AECPFYFQVRUWC3KGNLJIOREFP5LQ,AGYYVPDD7YG7FYNBX...
2 AGU3BBQ2V2DDAM0AKGFAWDDQ6QHA,AESFLDV2PT363T2AQ...
3 AEWAZDZZJLQUYV0VGBEUKSLXHQ5A,AG5HTSFRRE6NL3M5S...
4 AE3Q6KSUK5P75D5HFYHCRA0LODSA,AFUGIFH5ZAFXRDSZH...

user_name \

0 Manav,Adarsh gupta,Sundeep,S.Sayeed Ahmed,jasp...
1 ArdKn,Nirbhay kumar,Sagar Viswanathan,Asp,Plac...
2 Kunal,Himanshu,viswanath,sai niharka,saqib mal...
3 Omkar dhale,JD,HEMALATHA,Ajwadh a.,amar singh ...
4 rahuls6099,Swasat Borah,Ajay Wadke,Pranali,RVK...

review_id \

0 R3HXWT0LRP0NMF,R2AJM3LFTLZHF0,R6AQJGUP6P86,R1K...
1 RGIQEG07R9HS2,R1SMWZQ86XIN8U,R2J3Y1WL29GWDE,RY...
2 R3J3EQ09TZI5ZJ,R3E7WBGK7ID0KV,RWU79XKQ6I1QF,R2...
3 R3EEUZKKK9J36I,R3HJVYCLY0Y554,REDECAZ7AMPQC,R1...
4 R1BP4L2HH9TFUP,R16PVJEXKV6QZS,R2UPDB81N66T4P,R...

review_title \

0 Satisfied,Charging is really fast,Value for mo...
1 A Good Braided Cable for Your Type C Device,Go...
2 Good speed for earlier versions,Good Product,W...
3 Good product,Good one,Nice,Really nice product...
4 As good as original,Decent,Good one for second...

review_content \

0 Looks durable Charging is fine tooNo complains...
1 I ordered this cable to connect my phone to An...
2 Not quite durable and sturdy,https://m.media-a...
3 Good product,long wire,Charges good,Nice,I bou...
4 Bought this instead of original apple, does th...

img_link \

0 https://m.media-amazon.com/images/W/WEBP_40237...
1 https://m.media-amazon.com/images/W/WEBP_40237...
2 https://m.media-amazon.com/images/W/WEBP_40237...
3 https://m.media-amazon.com/images/I/41V5FtEWPk...
4 https://m.media-amazon.com/images/W/WEBP_40237...

product_link

0 https://www.amazon.in/Wayona-Braided-WN3LG1-Sy...
1 https://www.amazon.in/Ambrane-Unbreakable-Char...

```

2 https://www.amazon.in/Sounce-iPhone-Charging-C...
3 https://www.amazon.in/Deuce-300-Resistant-Tang...
4 https://www.amazon.in/Portronics-Konnect-POR-1...

```

```
df.tail()
```

	product_id	product_name	\
1460	B08L7J3T31	Noir Aqua - 5pcs PP Spun Filter + 1 Spanner	...
1461	B01M6453MB	Prestige Delight PRW0 Electric Rice Cooker (1 ...	
1462	B009P2LIL4	Bajaj Majesty RX10 2000 Watts Heat Convector R...	
1463	B00J5DYCCA	Havells Ventil Air DSP 230mm Exhaust Fan (Pist...	
1464	B01486F4G6	Borosil Jumbo 1000-Watt Grill Sandwich Maker (...)	

	category
discounted_price	\
1460	Home&Kitchen Kitchen&HomeAppliances WaterPurif...
	₹379
1461	Home&Kitchen Kitchen&HomeAppliances SmallKitch...
	₹2,280
1462	Home&Kitchen Heating,Cooling&AirQuality RoomHe...
	₹2,219
1463	Home&Kitchen Heating,Cooling&AirQuality Fans E...
	₹1,399
1464	Home&Kitchen Kitchen&HomeAppliances SmallKitch...
	₹2,863

	actual_price	discount_percentage	rating	rating_count	\
1460	₹919	59%	4	1,090	
1461	₹3,045	25%	4.1	4,118	
1462	₹3,080	28%	3.6	468	
1463	₹1,890	26%	4	8,031	
1464	₹3,690	22%	4.3	6,987	

	about_product	\
1460	SUPREME QUALITY 90 GRAM 3 LAYER THIK PP SPUN F...	
1461	230 Volts, 400 watts, 1 Year	
1462	International design and styling Two heat sett...	
1463	Fan sweep area: 230 MM ; Noise level: (40 - 45...	
1464	Brand-Borosil, Specification " 23V ~ 5Hz;1 W...	

	user_id	\
1460	AHITFY6AHALOF0H0ZE0C6XBP4FEA,AFRABB0DZJZQB6Z4U...	
1461	AFG5FM3NEM0L6BNFRV2NK5FNJCHQ,AGEINTRN6Z563RMLH...	
1462	AGVPWCMAHYQWJ0QKMUJN4DW3KM5Q,AF4Q3E66MY4SR7YQZ...	
1463	AF2JQCLSCY3QJATWUNNHUSVUPNQQ,AFDMLUXC5LS5RXDJS...	
1464	AFGW5PT3R6ZAVQR4Y5MWVAKBZAYA,AG7QNJ2SCS5VS5VYY...	

	user_name	\
1460	Prabha ds,Raghuram bk,Real Deal,Amazon Custome...	
1461	Manu Bhai,Naveenpittu,Evatira Sangma,JAGANNADH...	

```

1462 Nehal Desai,Danish Parwez,Amazon Customer,Amaz...
1463 Shubham Dubey,E.GURUBARAN,Mayank S.,eusuf khan...
1464 Rajib,Ajay B,Vikas Kahol,PARDEEP,Anindya Prama...

```

```

                                review_id \
1460 R3G3XFHPBFF0E8,R3C0BZCD32EIGW,R2EBVBCN9QPD9R,R...
1461 R3DDL2UPKQ2CK9,R2SYYU10ATVIU5,R1VM993161IYRW,R...
1462 R1TLRJVW4STY5I,R20455KRN493R1,R3Q5MVGBRIAS2G,R...
1463 R39Q2Y79MM9SWK,R3079BG1NIH6MB,R29A31ZELTZNJM,R...
1464 R20RBRZ0WEUJT9,R0KIFK9R2ISSE,R30EEG2FNJSN5I,R2...

```

```

                                review_title \
1460 Received the product without spanner,Excellent...
1461 ok,everything was good couldn't return bcoz I ...
1462 very good,Work but front melt after 2 month,Go...
1463 Fan Speed is slow,Good quality,Good product,go...
1464 Works perfect,Ok good product,Nice Product. Re...

```

```

                                review_content \
1460 I received product without spanner,Excellent p...
1461 ok,got everything as mentioned but the measuri...
1462 plastic but cool body ,u have to find sturdy s...
1463 I have installed this in my kitchen working fi...
1464 It does it job perfectly..only issue is temp c...

```

```

                                img_link \
1460 https://m.media-amazon.com/images/I/41fDdRtjfx...
1461 https://m.media-amazon.com/images/I/41gzDxk4+k...
1462 https://m.media-amazon.com/images/W/WEBP_40237...
1463 https://m.media-amazon.com/images/W/WEBP_40237...
1464 https://m.media-amazon.com/images/W/WEBP_40237...

```

```

                                product_link
1460 https://www.amazon.in/Noir-Aqua-Spanner-Purifi...
1461 https://www.amazon.in/Prestige-Delight-PRW0-1-...
1462 https://www.amazon.in/Bajaj-RX-10-2000-Watt-Co...
1463 https://www.amazon.in/Havells-Ventilair-230mm-...
1464 https://www.amazon.in/Borosil-Jumbo-1000-Watt-...

```

```
df.shape
```

```
(1465, 16)
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 1465 entries, 0 to 1464
```

```
Data columns (total 16 columns):
```

#	Column	Non-Null Count	Dtype
0	product_id	1465 non-null	object

1	product_name	1465	non-null	object
2	category	1465	non-null	object
3	discounted_price	1465	non-null	object
4	actual_price	1465	non-null	object
5	discount_percentage	1465	non-null	object
6	rating	1465	non-null	object
7	rating_count	1463	non-null	object
8	about_product	1465	non-null	object
9	user_id	1465	non-null	object
10	user_name	1465	non-null	object
11	review_id	1465	non-null	object
12	review_title	1465	non-null	object
13	review_content	1465	non-null	object
14	img_link	1465	non-null	object
15	product_link	1465	non-null	object

dtypes: object(16)

memory usage: 183.3+ KB

df.describe()

	product_id	product_name
count	1465	1465
unique	1351	1337
top	B07JW9H4J1	Fire-Boltt Ninja Call Pro Plus 1.83" Smart Wat...
freq	3	5

	category
discounted_price \	
count	1465
1465	
unique	211
550	
top	Computers&Accessories Accessories&Peripherals ...
₹199	
freq	233
53	

	actual_price	discount_percentage	rating	rating_count	\
count	1465	1465	1465	1463	
unique	449	92	28	1143	
top	₹999	50%	4.1	9,378	
freq	120	56	244	9	

	about_product	\
count	1465	

```

unique                                1293
top      [CHARGE & SYNC FUNCTION]- This cable comes wit...
freq                                           6

                                user_id \
count                                1465
unique                              1194
top      AHIKJUDTVJ4T6DV6IUGFYZ5LXMPA,AE55KTFVNXYPD5FPY...
freq                                           10

                                user_name \
count                                1465
unique                              1194
top      $@|\|T0$|-|,Sethu madhav,Akash Thakur,Burger P...
freq                                           10

                                review_id \
count                                1465
unique                              1194
top      R3F4T5TRYPTMIG,R3DQIEC603E7AY,R104Z15FD40PV5,R...
freq                                           10

                                review_title \
count                                1465
unique                              1194
top      Worked on iPhone 7 and didn't work on XR,Good ...
freq                                           10

                                review_content \
count                                1465
unique                              1212
top      I am not big on camera usage, personally. I wa...
freq                                           8

                                img_link \
count                                1465
unique                              1412
top      https://m.media-amazon.com/images/I/413sCRKobN...
freq                                           3

                                product_link
count                                1465
unique                              1465
top      https://www.amazon.in/Wayona-Braided-WN3LG1-Sy...
freq                                           1

```

Step 1: Removing ₹ and commas

```

df["discounted_price"] = df["discounted_price"].replace(['₹,'], '',
regex=True)

```

```

# Step 2: Converting to number
df["discounted_price"] = pd.to_numeric(df["discounted_price"],
errors="coerce")

# Step 3: Check
print(df["discounted_price"].dtype)

float64

# Check single column
df["rating"]

# Multiple columns
df[["product_name", "rating", "discounted_price"]]

# Save dataset
df.to_csv("Detail.csv", index=False)

# Select specific columns
df[["product_name", "discounted_price"]]

# Filter products with price > 500
df[df["discounted_price"] > 500][["product_name", "discounted_price"]]

```

```

                                product_name
discounted_price
16    MI 80 cm (32 inches) 5A Series HD Ready Smart ...
13999.0
19    LG 80 cm (32 inches) HD Ready Smart LED TV 32L...
13490.0
20    Duracell USB Lightning Apple Certified (Mfi) B...
970.0
22    Samsung 80 cm (32 Inches) Wondertainment Serie...
13490.0
24    Acer 80 cm (32 inches) I Series HD Ready Andro...
11499.0
...
...
1458  Libra Room Heater for Home, Room Heaters Home ...
949.0
1461  Prestige Delight PRW0 Electric Rice Cooker (1 ...
2280.0
1462  Bajaj Majesty RX10 2000 Watts Heat Convector R...
2219.0
1463  Havells Ventil Air DSP 230mm Exhaust Fan (Pist...
1399.0
1464  Borosil Jumbo 1000-Watt Grill Sandwich Maker (...
2863.0

```

```
[889 rows x 2 columns]
```

Insight:

Dataset contains product-related features such as price, rating, and category. Initial exploration helps understand structure and data types.

Step 4: Data Cleaning

1. Check Format

```
df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1465 entries, 0 to 1464
Data columns (total 16 columns):
#   Column                Non-Null Count  Dtype  
---  -
0   product_id            1465 non-null   object  
1   product_name          1465 non-null   object  
2   category              1465 non-null   object  
3   discounted_price       1465 non-null   float64  
4   actual_price          1465 non-null   object  
5   discount_percentage    1465 non-null   object  
6   rating                1465 non-null   object  
7   rating_count          1463 non-null   object  
8   about_product         1465 non-null   object  
9   user_id              1465 non-null   object  
10  user_name             1465 non-null   object  
11  review_id            1465 non-null   object  
12  review_title         1465 non-null   object  
13  review_content       1465 non-null   object  
14  img_link             1465 non-null   object  
15  product_link         1465 non-null   object  
dtypes: float64(1), object(15)
memory usage: 183.3+ KB
```

Insight:

Data types are checked to ensure correct analysis.

2. Check Null Values

```
df.isnull().sum()

product_id            0
product_name          0
category              0
discounted_price      0
actual_price          0
discount_percentage   0
rating               0
rating_count          2
```



```
about_product      0
user_id            0
user_name          0
review_id          0
review_title       0
review_content     0
img_link           0
product_link       0
dtype: int64
```

3. Impute Null Values

```
df["rating_count"] = df["rating_count"].fillna(0)

df.isnull().sum()
```

```
product_id      0
product_name    0
category        0
discounted_price 0
actual_price    0
discount_percentage 0
rating          0
rating_count    0
about_product   0
user_id         0
user_name       0
review_id       0
review_title    0
review_content  0
img_link        0
product_link    0
dtype: int64
```

Mean:

```
df["rating_count"].dtype
dtype('O')

df["rating_count"] = pd.to_numeric(df["rating_count"],
errors="coerce")

df["rating_count"] =
df["rating_count"].fillna(df["rating_count"].mean())
```

Median:

```
df["rating_count"] =
df["rating_count"].fillna(df["rating_count"].median(),inplace=True)
```

```
C:\Users\masto\AppData\Local\Temp\ipykernel_11308\688903511.py:1:
FutureWarning: A value is trying to be set on a copy of a DataFrame or
Series through chained assignment using an inplace method.
The behavior will change in pandas 3.0. This inplace method will never
work because the intermediate object on which we are setting values
always behaves as a copy.
```

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col] = df[col].method(value) instead, to perform the operation inplace on the original object.

```
df["rating_count"] =
df["rating_count"].fillna(df["rating_count"].median(),inplace=True)
```

Mode:

```
df["rating_count"] =
df["rating_count"].fillna(df["rating_count"].mode(),inplace=True)
```

```
C:\Users\masto\AppData\Local\Temp\ipykernel_11308\2278854782.py:1:
FutureWarning: A value is trying to be set on a copy of a DataFrame or
Series through chained assignment using an inplace method.
The behavior will change in pandas 3.0. This inplace method will never
work because the intermediate object on which we are setting values
always behaves as a copy.
```

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col] = df[col].method(value) instead, to perform the operation inplace on the original object.

```
df["rating_count"] =
df["rating_count"].fillna(df["rating_count"].mode(),inplace=True)
```

Insight:

Missing values were handled using statistical techniques to maintain data quality.

4. Check Duplicates

```
df.duplicated()
df.duplicated().sum()

np.int64(0)
```

Remove:

```
df.drop_duplicates(inplace=True)
```

Insight:

Duplicate records were removed to avoid bias in analysis.

5. Drop Rows / Columns

```
df.drop(0, axis=0) # row delete
df.drop("img_link", axis=1) # column delete
```

	product_id	product_name	\
0	B07JW9H4J1	Wayona Nylon Braided USB to Lightning Fast Cha...	
1	B098NS6PVG	Ambrane Unbreakable 60W / 3A Fast Charging 1.5...	
2	B096MSW6CT	Sounce Fast Phone Charging Cable & Data Sync U...	
3	B08HDJ86NZ	boAt Deuce USB 300 2 in 1 Type-C & Micro USB S...	
4	B08CF3B7N1	Portronics Konnect L 1.2M Fast Charging 3A 8 P...	
...	...	...	...
1460	B08L7J3T31	Noir Aqua - 5pcs PP Spun Filter + 1 Spanner ...	
1461	B01M6453MB	Prestige Delight PRW0 Electric Rice Cooker (1 ...	
1462	B009P2LIL4	Bajaj Majesty RX10 2000 Watts Heat Convector R...	
1463	B00J5DYCCA	Havells Ventil Air DSP 230mm Exhaust Fan (Pist...	
1464	B01486F4G6	Borosil Jumbo 1000-Watt Grill Sandwich Maker (...	

	category
discounted_price \	
0	Computers&Accessories Accessories&Peripherals ...
399.0	
1	Computers&Accessories Accessories&Peripherals ...
199.0	
2	Computers&Accessories Accessories&Peripherals ...
199.0	
3	Computers&Accessories Accessories&Peripherals ...
329.0	
4	Computers&Accessories Accessories&Peripherals ...
154.0	
...	...
...	...
1460	Home&Kitchen Kitchen&HomeAppliances WaterPurif...
379.0	
1461	Home&Kitchen Kitchen&HomeAppliances SmallKitch...
2280.0	
1462	Home&Kitchen Heating,Cooling&AirQuality RoomHe...
2219.0	
1463	Home&Kitchen Heating,Cooling&AirQuality Fans E...
1399.0	
1464	Home&Kitchen Kitchen&HomeAppliances SmallKitch...
2863.0	

	actual_price	discount_percentage	rating	rating_count	\
0	₹1,099	64%	4.2	None	
1	₹349	43%	4.0	None	
2	₹1,899	90%	3.9	None	

3	₹699	53%	4.2	None
4	₹399	61%	4.2	None
...	...	...	...	...
1460	₹919	59%	4	None
1461	₹3,045	25%	4.1	None
1462	₹3,080	28%	3.6	None
1463	₹1,890	26%	4	None
1464	₹3,690	22%	4.3	None

	about_product \
0	High Compatibility : Compatible With iPhone 12...
1	Compatible with all Type C enabled devices, be...
2	【Fast Charger& Data Sync】-With built-in safet...
3	The boAt Deuce USB 300 2 in 1 cable is compati...
4	[CHARGE & SYNC FUNCTION]- This cable comes wit...
...	...
1460	SUPREME QUALITY 90 GRAM 3 LAYER THIK PP SPUN F...
1461	230 Volts, 400 watts, 1 Year
1462	International design and styling Two heat sett...
1463	Fan sweep area: 230 MM ; Noise level: (40 - 45...
1464	Brand-Borosil, Specification â€” 23V ~ 5Hz;1 W...

	user_id \
0	AG3D604STAQKAY2UVGEUV46KN35Q,AHMY5CWJMMK5BJRBB...
1	AECPFYFQVRUWC3KGNLJIOREFP5LQ,AGYYVPDD7YG7FYNBX...
2	AGU3BBQ2V2DDAMOAKGFAWDDQ6QHA,AESFLDV2PT363T2AQ...
3	AEWAZDZZJLQUYV0VGBEUKSLXHQ5A,AG5HTSFRRE6NL3M5S...
4	AE3Q6KSUK5P75D5HFYHCRA0LODSA,AFUGIFH5ZAFXRDSZH...
...	...
1460	AHITFY6AHAL0F0H0ZE0C6XBP4FEA,AFRABB0DZJZQB6Z4U...
1461	AFG5FM3NEMOL6BNFRV2NK5FNJCHQ,AGEINTRN6Z563RMLH...
1462	AGVPWCMAHYQWJOQKMUJN4DW3KM5Q,AF4Q3E66MY4SR7YQZ...
1463	AF2JQCLSCY3QJATWUNNHUSVUPNQQ,AFDMLUXC5LS5RXDJS...
1464	AFGW5PT3R6ZAVQR4Y5MWVAKBZAYA,AG7QNJ2SCS5VS5VYY...

	user_name \
0	Manav,Adarsh gupta,Sundeep,S.Sayeed Ahmed,jasp...
1	ArdKn,Nirbhay kumar,Sagar Viswanathan,Asp,Plac...
2	Kunal,Himanshu,viswanath,sai niharka,saqib mal...
3	Omkar dhale,JD,HEMALATHA,Ajwadh a.,amar singh ...
4	rahuls6099,Swasat Borah,Ajay Wadke,Pranali,RVK...
...	...
1460	Prabha ds,Raghuram bk,Real Deal,Amazon Custome...
1461	Manu Bhai,Naveenpittu,Evatira Sangma,JAGANNADH...
1462	Nehal Desai,Danish Parwez,Amazon Customer,Amaz...
1463	Shubham Dubey,E.GURUBARAN,Mayank S.,eusuf khan...
1464	Rajib,Ajay B,Vikas Kahol,PARDEEP,Anindya Prama...

	review_id \
0	R3HXWT0LRP0NMF,R2AJM3LFTLZHF0,R6AQJGUP6P86,R1K...

```

1      RGIQEG07R9HS2,R1SMWZQ86XIN8U,R2J3Y1WL29GWDE,RY...
2      R3J3EQQ9TZI5ZJ,R3E7WBGK7ID0KV,RWU79XKQ6I1QF,R2...
3      R3EEUZKKK9J36I,R3HJVYCLY0Y554,REDECAZ7AMPQC,R1...
4      R1BP4L2HH9TFUP,R16PVJEXKV6QZS,R2UPDB81N66T4P,R...
...
1460   R3G3XFHPBFF0E8,R3C0BZCD32EIGW,R2EBVBCN9QPD9R,R...
1461   R3DDL2UPKQ2CK9,R2SYYU10ATVIU5,R1VM993161IYRW,R...
1462   R1TLRJVW4STY5I,R20455KRN493R1,R3Q5MVGBRIAS2G,R...
1463   R39Q2Y79MM9SWK,R3079BG1NIH6MB,R29A31ZELTZNJM,R...
1464   R20RBRZ0WEUJT9,R0KIFK9R2ISSE,R30EEG2FNJSN5I,R2...

```

review_title \

```

0      Satisfied,Charging is really fast,Value for mo...
1      A Good Braided Cable for Your Type C Device,Go...
2      Good speed for earlier versions,Good Product,W...
3      Good product,Good one,Nice,Really nice product...
4      As good as original,Decent,Good one for second...
...
1460   Received the product without spanner,Excellent...
1461   ok,everything was good couldn't return bcoz I ...
1462   very good,Work but front melt after 2 month,Go...
1463   Fan Speed is slow,Good quality,Good product,go...
1464   Works perfect,Ok good product,Nice Product. Re...

```

review_content \

```

0      Looks durable Charging is fine tooNo complains...
1      I ordered this cable to connect my phone to An...
2      Not quite durable and sturdy,https://m.media-a...
3      Good product,long wire,Charges good,Nice,I bou...
4      Bought this instead of original apple, does th...
...
1460   I received product without spanner,Excellent p...
1461   ok,got everything as mentioned but the measuri...
1462   plastic but cool body ,u have to find sturdy s...
1463   I have installed this in my kitchen working fi...
1464   It does it job perfectly..only issue is temp c...

```

product_link

```

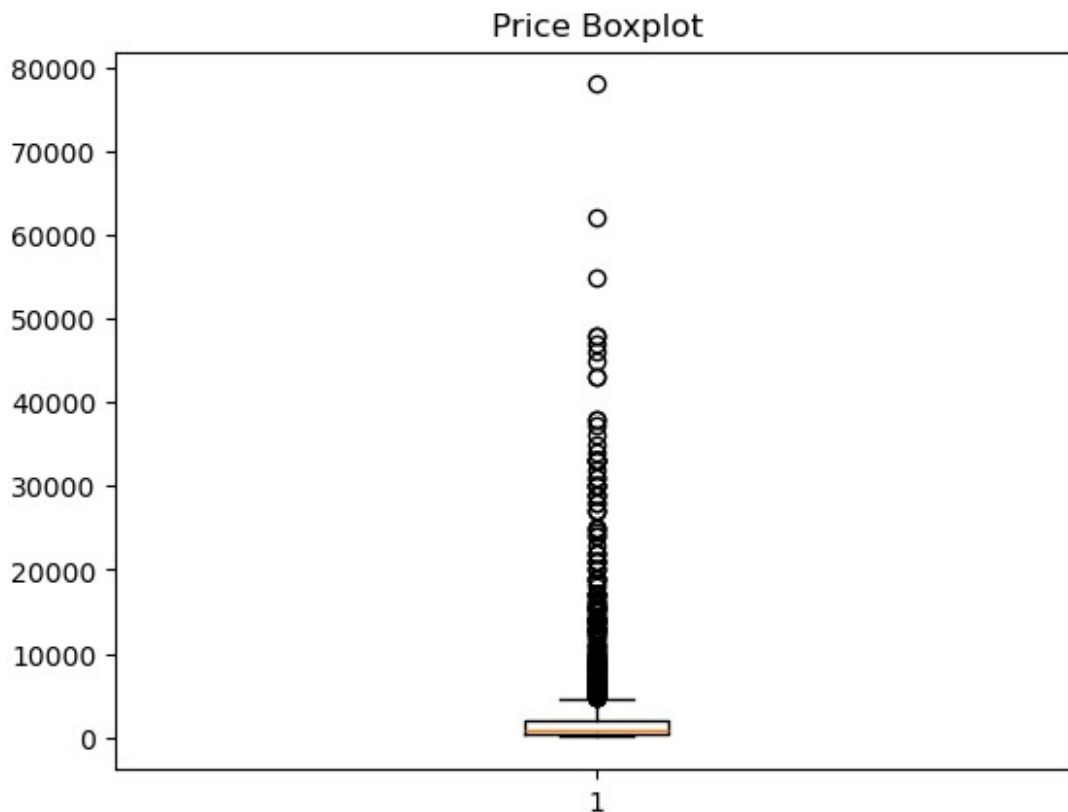
0      https://www.amazon.in/Wayona-Braided-WN3LG1-Sy...
1      https://www.amazon.in/Ambrane-Unbreakable-Char...
2      https://www.amazon.in/Sounce-iPhone-Charging-C...
3      https://www.amazon.in/Deuce-300-Resistant-Tang...
4      https://www.amazon.in/Portronics-Konnect-POR-1...
...
1460   https://www.amazon.in/Noir-Aqua-Spanner-Purifi...
1461   https://www.amazon.in/Prestige-Delight-PRW0-1-...
1462   https://www.amazon.in/Bajaj-RX-10-2000-Watt-Co...
1463   https://www.amazon.in/Havells-Ventilair-230mm-...
1464   https://www.amazon.in/Borosil-Jumbo-1000-Watt-...

```

[1465 rows x 15 columns]

6. Check Outliers

```
plt.boxplot(df["discounted_price"])  
plt.title("Price Boxplot")  
plt.show()
```



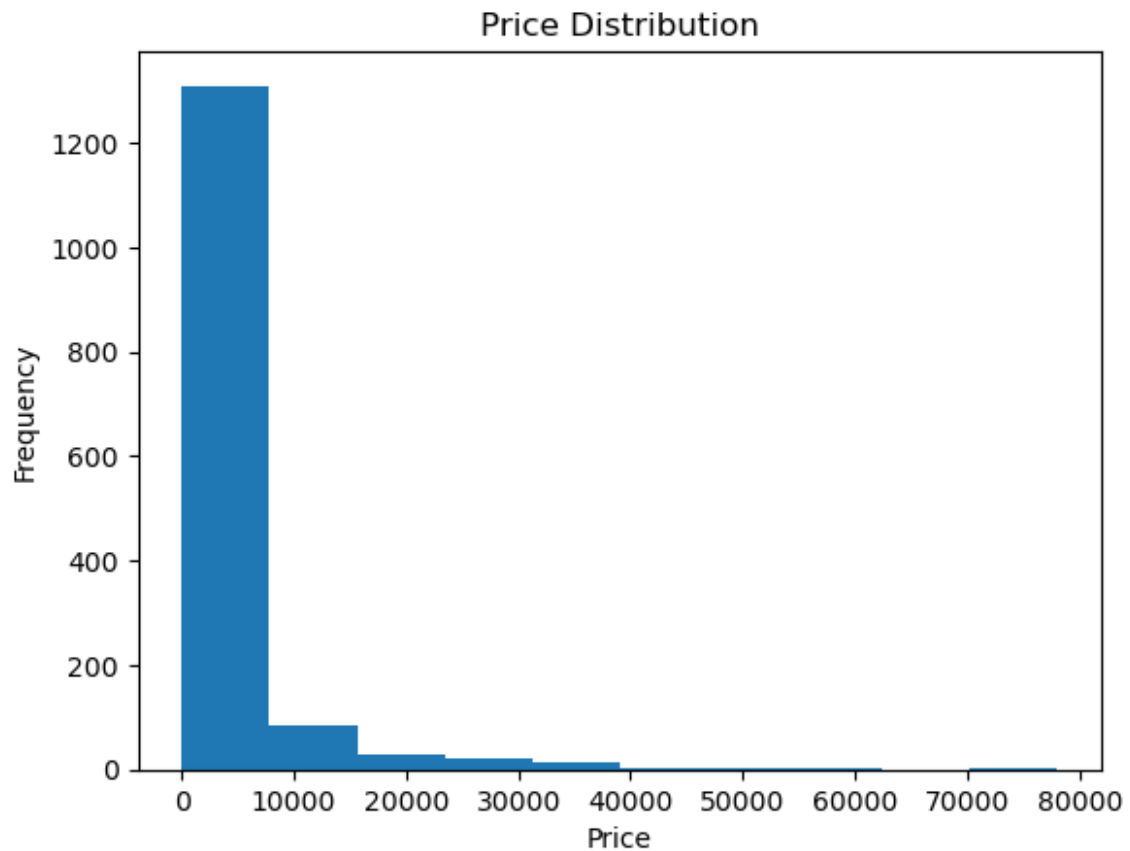
Insight:

Outliers were detected in price, indicating presence of very high-value products.

Step 5: EDA (Exploratory Data Analysis)

1. Univariate Analysis

```
plt.hist(df["discounted_price"])  
plt.title("Price Distribution")  
plt.xlabel("Price")  
plt.ylabel("Frequency")  
plt.show()
```

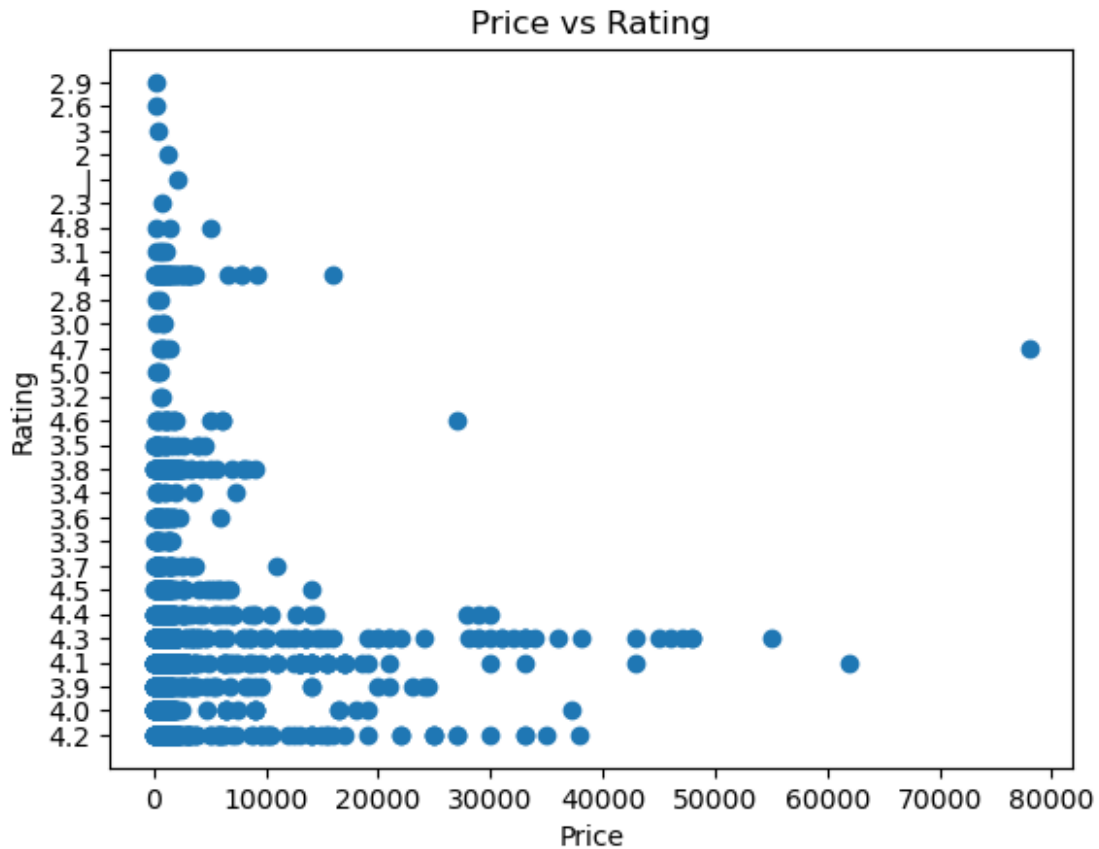


Insight:

Most products are in lower price range, with few high-priced items.

2. Bivariate Analysis

```
plt.scatter(df["discounted_price"], df["rating"])
plt.xlabel("Price")
plt.ylabel("Rating")
plt.title("Price vs Rating")
plt.show()
```

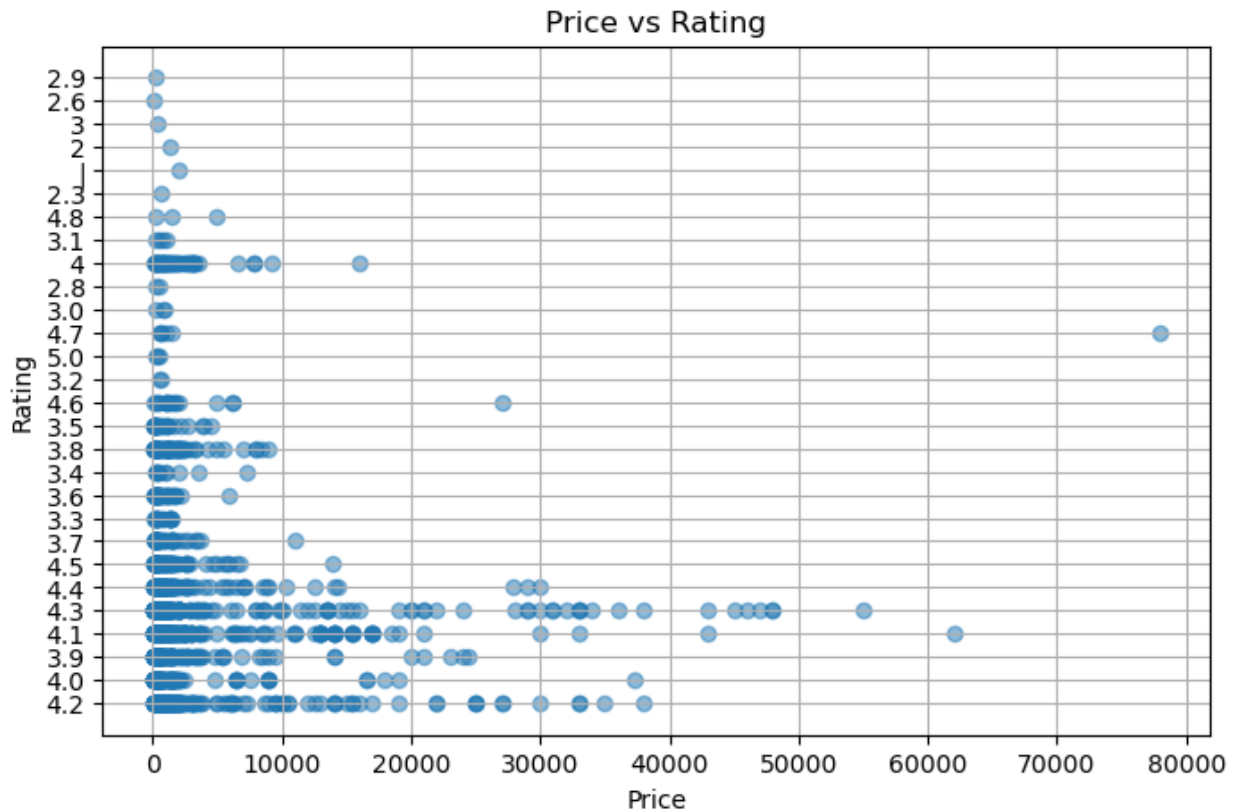


Insight:

No strong relationship between price and rating. Expensive products are not always highly rated.

3. Multivariate Analysis

```
plt.figure(figsize=(8,5))  
plt.scatter(df["discounted_price"], df["rating"], alpha=0.5)  
plt.xlabel("Price")  
plt.ylabel("Rating")  
plt.title("Price vs Rating")  
plt.grid()  
plt.show()
```

Insight:

A scatter plot was used to analyze the relationship between product price and rating.

The visualization shows that there is no strong correlation between price and rating.

Both low-priced and high-priced products have a wide range of ratings.

Category Analysis

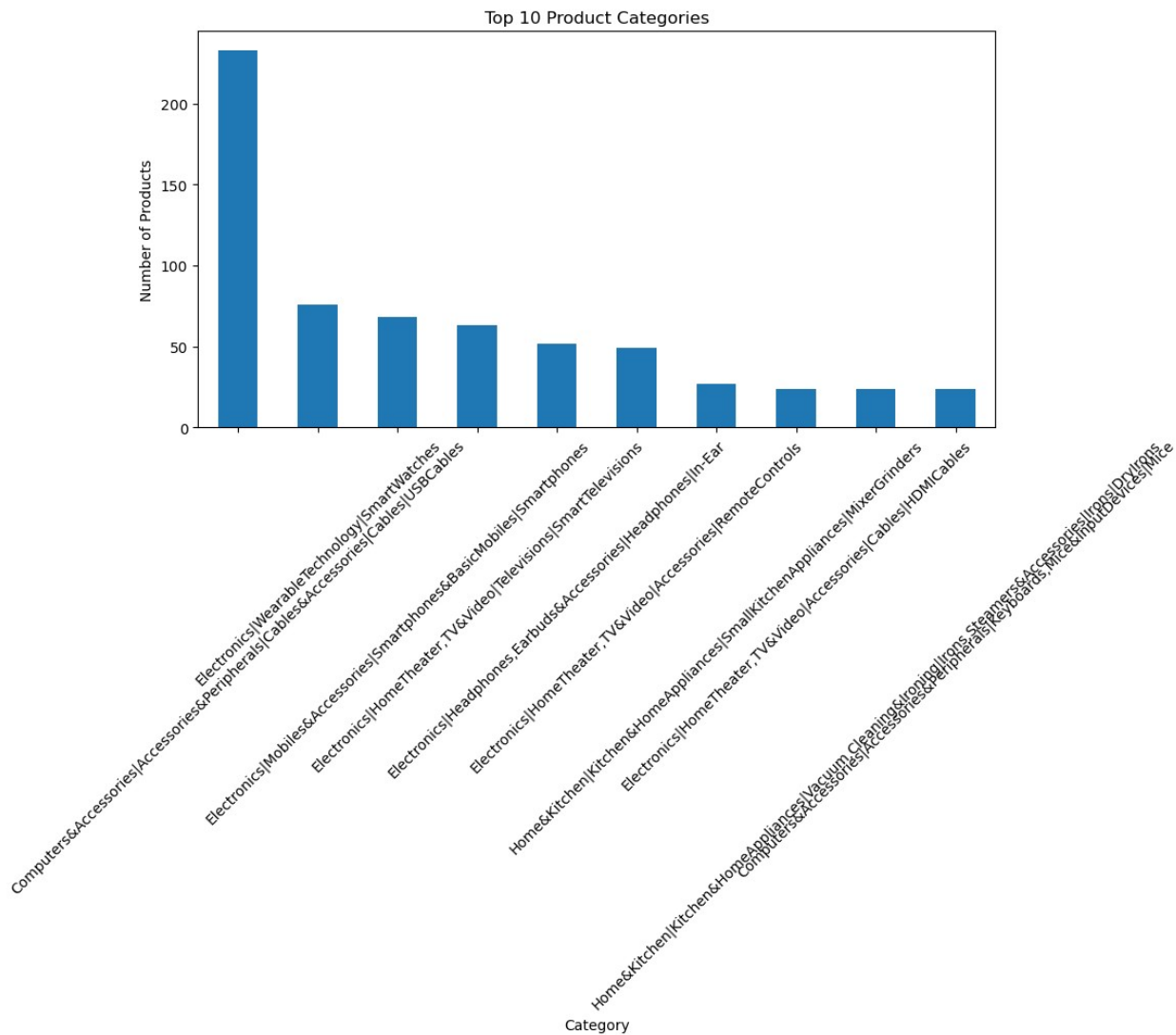
```
plt.figure(figsize=(10,5))

df["category"].value_counts().head(10).plot(kind="bar")

plt.title("Top 10 Product Categories")
plt.xlabel("Category")
plt.ylabel("Number of Products")

plt.xticks(rotation=45)

plt.show()
```



A bar chart was used to analyze the distribution of products across different categories.

The results show that a few categories contain a significantly higher number of products compared to others.

This indicates that certain categories dominate the platform, reflecting higher product availability and possibly greater consumer demand.

Final Conclusion

This project analyzes Amazon product data using Pandas and Matplotlib.

Data cleaning was performed to handle missing values, duplicates, and inconsistent formats in price-related columns.

Exploratory Data Analysis revealed that most products are priced in the lower to mid range, while a few products are significantly more expensive, indicating the presence of outliers.

Product ratings are generally high, suggesting positive customer satisfaction across the platform.

The analysis also showed that there is no strong correlation between product price and rating, meaning higher-priced products are not necessarily better rated.

Additionally, category-based analysis revealed that certain product categories dominate the dataset, indicating higher availability and potential demand in those segments.

These insights can help businesses understand pricing patterns, customer preferences, and category-level competition.